

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		80°C (1) because {the cells are ruptured/ membrane protein denatures} and (all) the pigment has leaked out of the cells/ (1)		2		2		2
		(ii)		All plots correct = 2 marks 1 plot incorrect = 1 mark >1 plot incorrect = 0 marks ½ small square tolerance All points plotted correctly with a smooth curve drawn or a dot to dot plot (1) No extrapolation with plot to plot /small amount allowed with line of best fit		3		3	3	
		(iii)		(The absorbance is not directly proportional to the temperature) {as it is not a straight line/ graph is a curve/ a description of changes in gradient} (1)			1	1		1
		(iv)		Any two (x1) from: (Up to 40°C) As temperature rises the {membrane becomes more fluid/phospholipids gain more kinetic energy/ more gaps form} (1) (Above 40°C) the proteins in the membrane start to denature (1) (Above 70°C) membrane fully destroyed (so all pigment released) (1) reject reference to membrane/ phospholipids denaturing.		2		2		2